Configuration and provenance

Branch: `feat/endtop-sslg4`
Commits: `0201413`, `7b4bc80`, `ad26d78`, `bcefa24`, `bdfe955`; this deck is generated from those published sources

- 31 positions $\times$ 2000 events; 16 End + 70 Top channels; jitter fixed to zero.
- EJ-204 via SSLG4 `OPSC-101`: yield 10400/MeV, $\tau_r = 0.7$ ns, $\tau_d = 1.8$ ns, ABSLENGTH=160 cm (verified effective MPT inputs).
- Broadcom AFBR-S4N66P024M: PDE 63% at 420 nm.

Scope

Top 70 SiPM is a simulation coverage study. Real hardware: 32 Top SiPMs. All reported timing is intrinsic, without SPTR or electronics jitter.

EndTop geometry and channel map

- EJ-204 bar: $1400 \times 60 \times 10$ mm^3.
- End L IDs 0–7; End R IDs 8–15.
- Top L IDs 16–50: $-692, -672, \ldots, -12$ mm.
- Top R IDs 51–85: $+12, +32, \ldots, +692$ mm.
- Each Top element and window: 6×6 mm^2.

Exact constraints

First centers are 5 mm from each bar end. Pitch is 20 mm except the deliberate central $-12/+12$ mm pair, whose pitch is 24 mm.

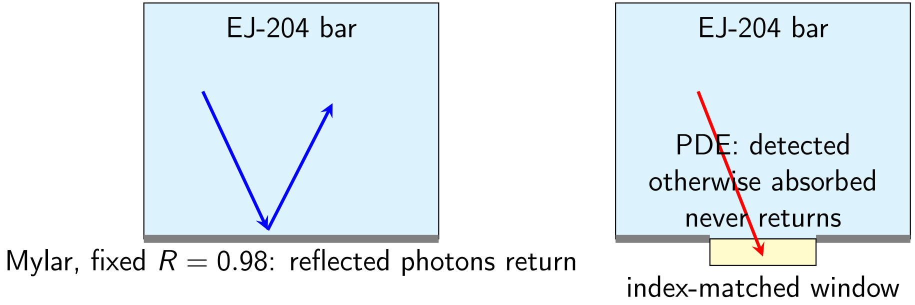
Instrumented and wrapped faces

Face	Optical state	Readout
$-X$, $+X$	open	8 End SiPMs each
$+Y$	Mylar with 70 exact windows	70 Top SiPMs
$-Y$, $-Z$, $+Z$	complete Mylar	none

External layer

Black Tedlar is deliberately not modeled; it only blocks ambient light. The non-reflected Mylar fraction is already absorbed in the optical model.

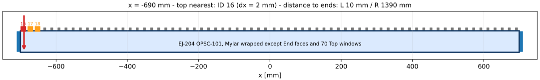
Window optics: why timing and collection change

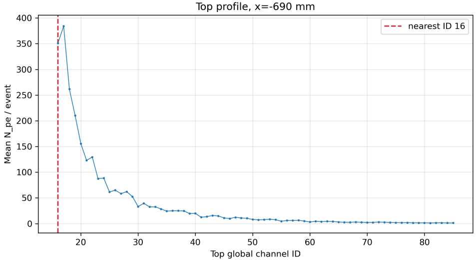


Consequence

Every window is an optical drain. Long, multiply reflected paths have more chances to meet one, so late End photons are preferentially removed.

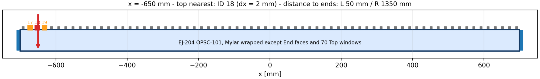
Position x=-690 mm

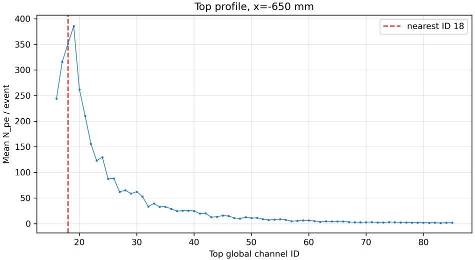




Quantity	Value
End L N_{pe}	3363.1 ± 18.1
End R N_{pe}	41.1 ± 0.3
Nearest Top ID	16 (window-track exception)
Nearest Top N_{pe}	350.9 ± 2.0
Nearest Top $\langle t \rangle$	2.7863 ± 0.0023 ns
Nearest Top FPT	0.34278 ± 0.00023 ns

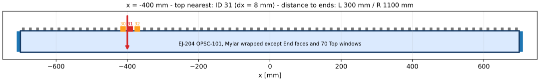
Position x=-650 mm

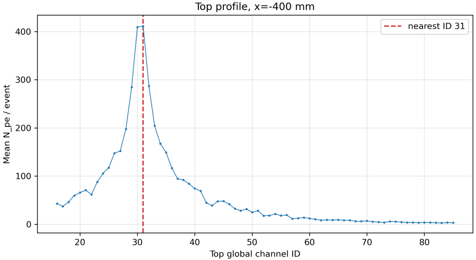




Quantity	Value
End L N_{pe}	2366.2 ± 13.1
End R N_{pe}	44.3 ± 0.3
Nearest Top ID	18 (window-track exception)
Nearest Top N_{pe}	351.0 ± 2.0
Nearest Top $\langle t \rangle$	2.7779 ± 0.0023 ns
Nearest Top FPT	0.34270 ± 0.00024 ns

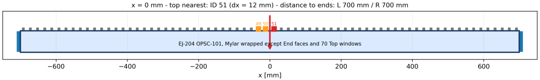
Position x=-400 mm

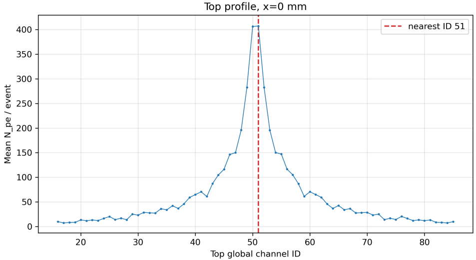




Quantity	Value
End L N_{pe}	691.3 ± 3.9
End R N_{pe}	74.5 ± 0.5
Nearest Top ID	31 (exact geometry)
Nearest Top N_{pe}	411.3 ± 2.3
Nearest Top $\langle t \rangle$	2.9399 ± 0.0022 ns
Nearest Top FPT	0.34597 ± 0.00021 ns

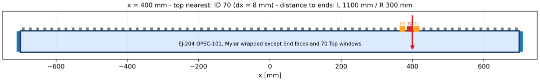
Position x=0 mm

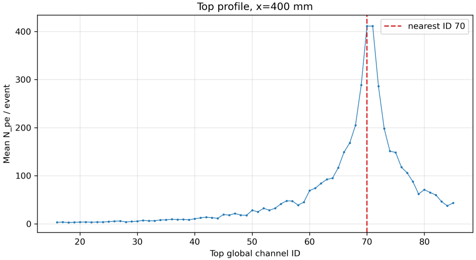




Quantity	Value
End L N_{pe}	194.4 ± 1.1
End R N_{pe}	195.2 ± 1.1
Nearest Top ID	51 (exact geometry)
Nearest Top N_{pe}	407.3 ± 2.2
Nearest Top $\langle t \rangle$	3.0030 ± 0.0022 ns
Nearest Top FPT	0.35075 ± 0.00020 ns

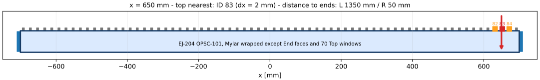
Position x=400 mm

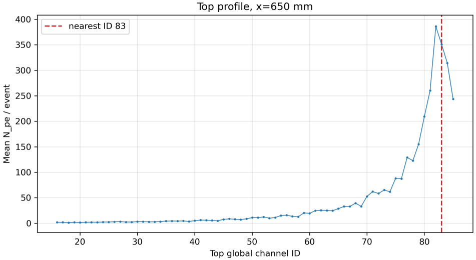




Quantity	Value
End L N_{pe}	75.0 ± 0.5
End R N_{pe}	692.5 ± 4.0
Nearest Top ID	70 (exact geometry)
Nearest Top N_{pe}	411.4 ± 2.4
Nearest Top $\langle t \rangle$	2.9433 ± 0.0022 ns
Nearest Top FPT	0.34598 ± 0.00022 ns

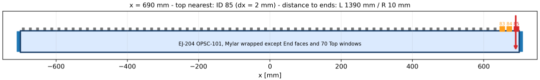
Position x=650 mm

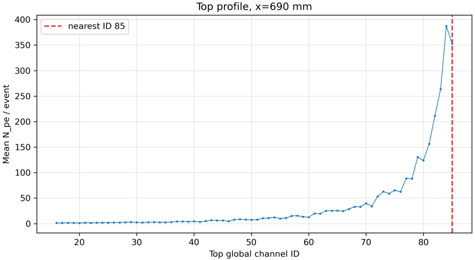




Quantity	Value
End L N_{pe}	44.4 ± 0.3
End R N_{pe}	2357.1 ± 13.0
Nearest Top ID	83 (window-track exception)
Nearest Top N_{pe}	350.9 ± 2.1
Nearest Top $\langle t \rangle$	2.7803 ± 0.0023 ns
Nearest Top FPT	0.34268 ± 0.00021 ns

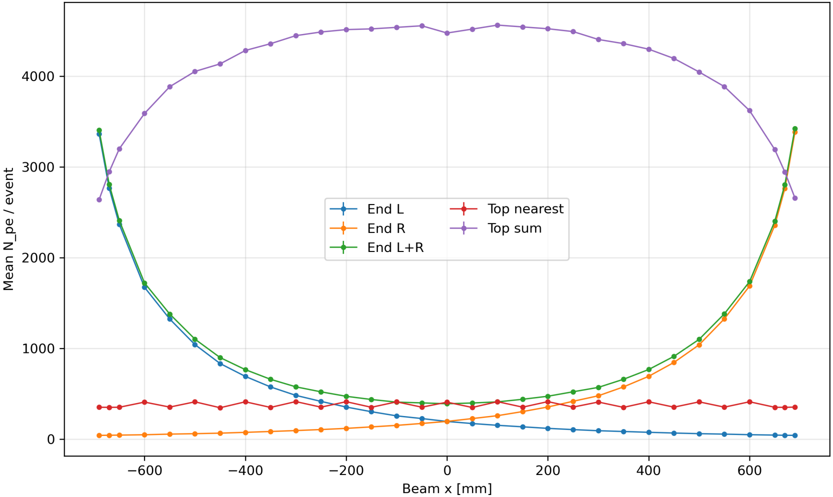
Position x=690 mm



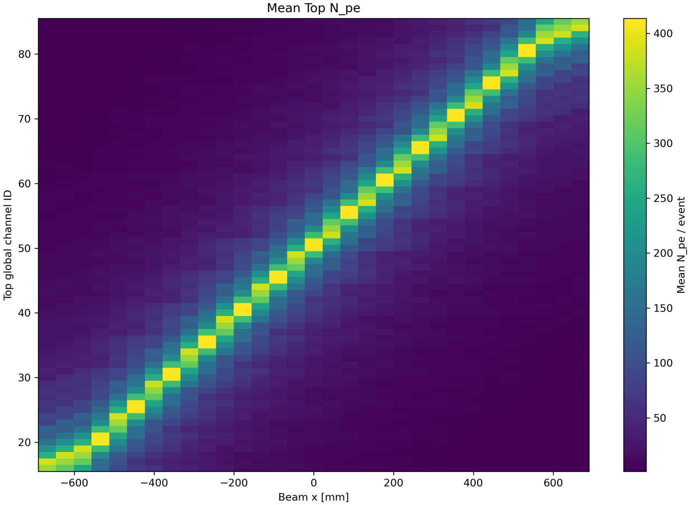


Quantity	Value
End L N_{pe}	41.4 ± 0.3
End R N_{pe}	3383.2 ± 18.3
Nearest Top ID	85 (window-track exception)
Nearest Top N_{pe}	353.0 ± 2.0
Nearest Top $\langle t \rangle$	2.7881 ± 0.0023 ns
Nearest Top FPT	0.34236 ± 0.00019 ns

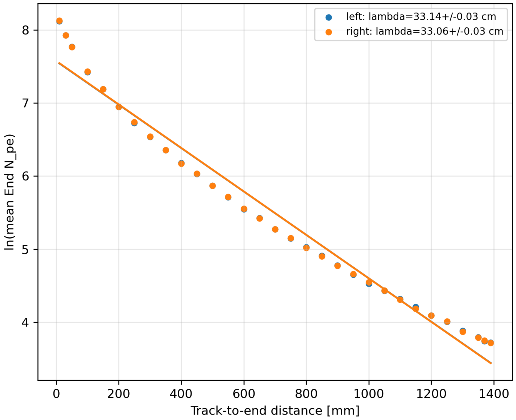
Photon budget versus beam position

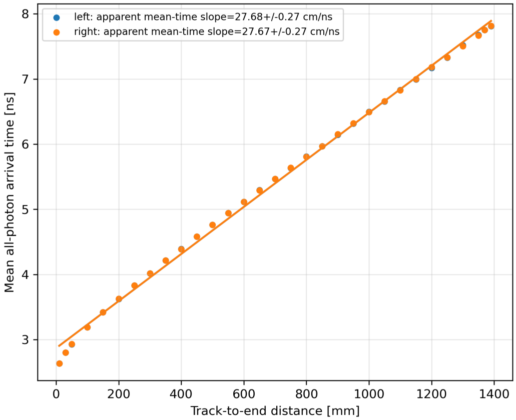


Top localization: full coverage diagonal



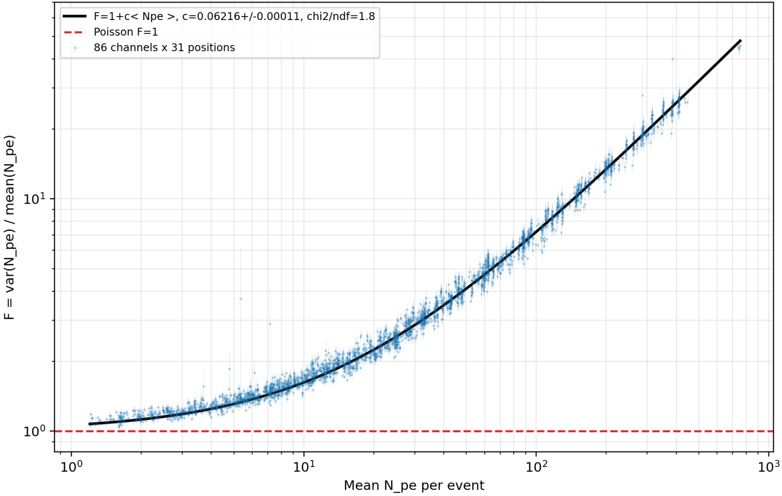
Attenuation and the historical mean-time slope





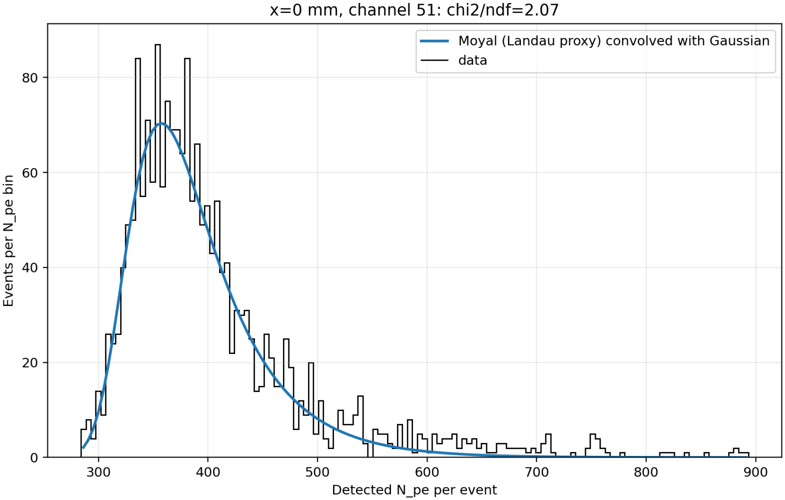
$\lambda_{eff,L} = 33.14 \pm 0.03$ cm, $\lambda_{eff,R} = 33.06 \pm 0.03$ cm; the mean-time slope is tail-biased and is not a propagation velocity.

Landau dominance: Fano factor versus mean Npe



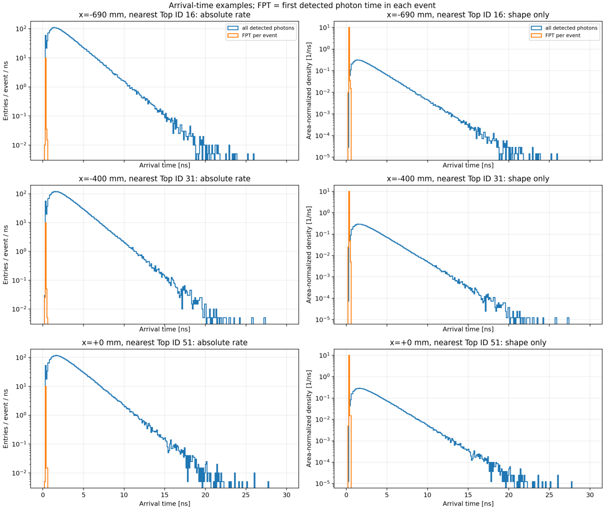
$F = 1 + c\langle N_{pe}\rangle$, $c = 0.062162 \pm 0.000107$, $\chi^2/ndf = 1.76$. Poisson $F = 1$ is approached only at low light.

High-light example: Moyal-Gaussian Landau proxy



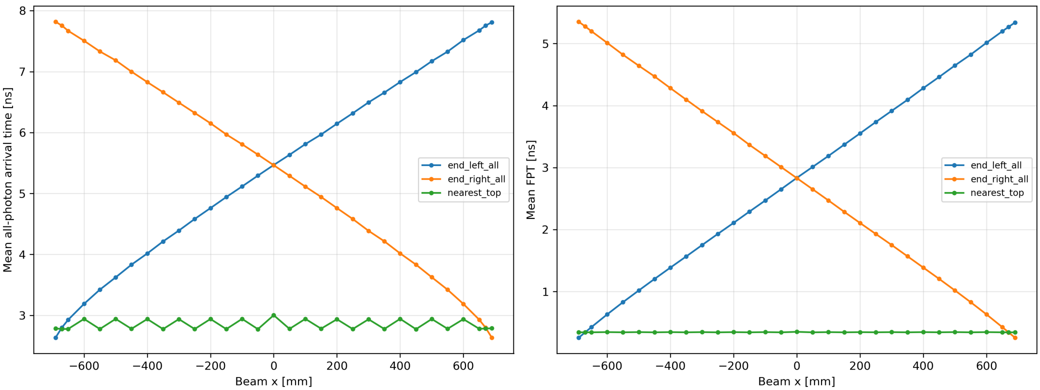
Moyal approximates Landau. Some channels have poor chi2 or an unidentifiable Gaussian width; all are recorded in `exec10_landau_mpv.csv`.

Arrival-time examples: absolute rate and normalized shape



Left: entries per event per ns. Right: the same distributions normalized to area 1. FPT is the first detected photon time in each event.

Mean arrival time and FPT versus x



Nearest-Top Landau/Moyal MPVs at key positions

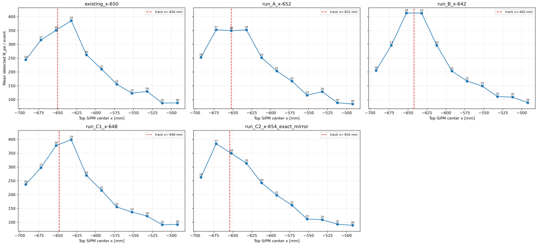
x [mm]	channel	MPV [Npe]	Landau width [Npe]	χ^2/ndf
-690	16	304.4 ± 1.0	26.9 ± 0.9	1.76
-650	18	302.4 ± 1.0	26.3 ± 0.9	1.04
-400	31	356.9 ± 1.1	29.3 ± 1.0	1.33
+0	51	355.4 ± 1.1	26.7 ± 0.9	2.07
+400	70	356.4 ± 1.2	29.1 ± 1.0	1.51
+650	83	304.1 ± 1.0	26.6 ± 0.9	1.61
+690	85	304.7 ± 1.0	25.6 ± 0.9	1.31

Interpretation

The right tail originates primarily from Landau-like energy-loss fluctuations; Poisson counting dominates only in the few-PE limit. MPVs are future inputs for the pending test-beam ToT(NPE) calibration comparison.

Window-track collection effect: five-run test

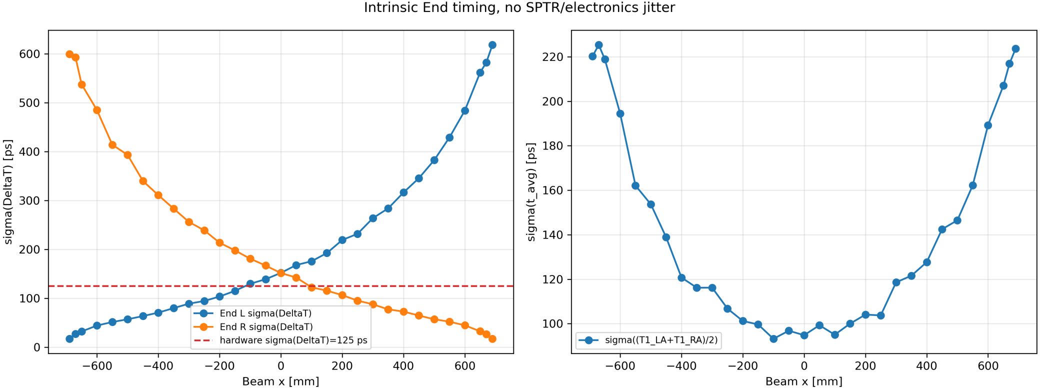
Run	contrast [Npe]	significance
Existing -650	34.60 ± 2.94	11.78
A -652 center	2.86 ± 2.65	1.08
B -642 midpoint	-0.33 ± 3.35	-0.10
C1 -648 (+4 mm)	21.08 ± 3.20	6.59
C2 -654 mirror	34.38 ± 2.95	11.64



Finding

Local symmetric ± 2 mm effect: about 9.8% at about 12 sigma; null when centered and at midpoint. The scan and channel map remain valid.

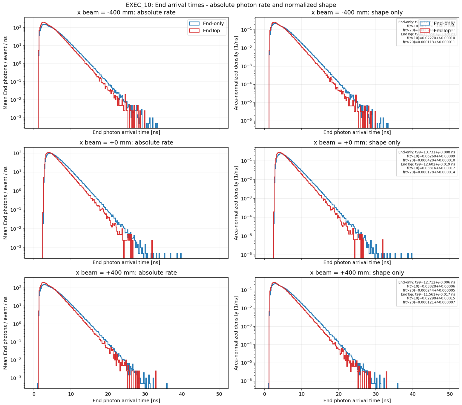
Intrinsic End SUM4 timing



$\sigma_{group} = \sigma(\Delta T_{AB})/\sqrt{2}$; no SPTR/jitter. The 88 ps hardware context is not directly comparable.

Timing gate lifted: narrower EndTop is physical

x	side	EndTop [ps]	End-only [ps]	ratio
-400	L	50.0 ± 0.8	66.1 ± 0.5	0.756 ± 0.013
-400	R	219.9 ± 3.5	248.4 ± 1.8	0.885 ± 0.015
+0	L	107.5 ± 1.7	131.4 ± 0.9	0.818 ± 0.014
+0	R	107.2 ± 1.7	129.5 ± 0.9	0.828 ± 0.014
+400	L	223.8 ± 3.5	247.5 ± 1.8	0.904 ± 0.016
+400	R	51.5 ± 0.8	65.9 ± 0.5	0.781 ± 0.014

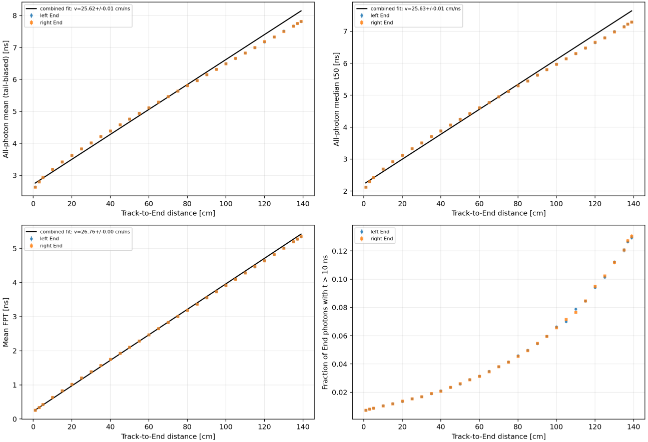


Confirmed mechanism

Across $x = 0, \pm 400$ mm, EndTop lowers $t99$ by 1.14 ns and suppresses late fractions at more than 10.3 sigma.

Left panels show absolute End photons/event/ns. Right panels normalize each distribution to area 1 and isolate shape: the $t > 10$ ns tail is suppressed in EndTop because multiply reflected photons have more opportunities to meet a Top window.

Apparent timing slopes: no propagation estimator passes



Estimator	slope-derived [cm/ns]	χ^2/ndf
all-photon mean	25.616 ± 0.005	3189.9
all-photon t_{50}	25.625 ± 0.005	3615.1
mean FPT	26.758 ± 0.003	1002.6

Refuted predictions

Mean FPT is not close to $c/n \simeq 19$ cm/ns. The fixed-threshold $f(t > 10\,ns)$ increases with distance. Every global line fit is rejected.

Top readout: analytic timing estimate only

x [mm]	nearest Top N_{pe}	$\sigma_{t,est}$ [ps]
-690	350.9 ± 2.0	59.92 ± 0.17
-650	351.0 ± 2.0	59.91 ± 0.17
-400	411.3 ± 2.3	55.35 ± 0.16
+0	407.3 ± 2.2	55.62 ± 0.15
+400	411.4 ± 2.4	55.34 ± 0.16
+650	350.9 ± 2.1	59.93 ± 0.18
+690	353.0 ± 2.0	59.75 ± 0.17

Definition and scope

$\sigma_{t,est} = \sqrt{0.7\,\mathrm{ns} \times 1.8\,\mathrm{ns}}/\sqrt{N_{pe}} = 1.122\,\mathrm{ns}/\sqrt{N_{pe}}$. This is an analytic orientation, not a simulated timing resolution; Top has no test-beam counterpart.

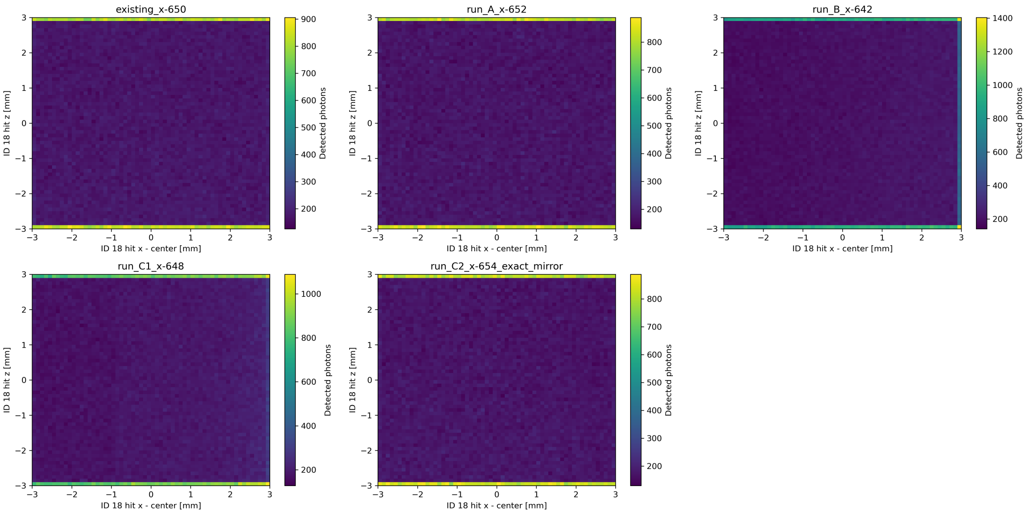
Numerical conclusions

- At center: End sum 389.6 ± 2.1, nearest Top 407.3 ± 2.2; at $x = -690$ mm: End sum 3404.2 ± 18.3, nearest Top 350.9 ± 2.0.
- λ_{eff}: L 33.14 ± 0.03 cm, R 33.06 ± 0.03 cm; no global time estimator yields an acceptable propagation-velocity fit.
- Nearest-Top var/mean: median 24.81, range 19.14–29.27; $F = 1 + (0.062162 \pm 0.000107)\langle N_{pe}\rangle$ with $\chi^2/ndf = 1.76$: Landau-like energy-loss fluctuations dominate.
- Intrinsic End σ_{group}: mean 148.3 ps, range 12.1–437.3 ps.
- EndTop/End-only timing ratio: 0.756 ± 0.016 to 0.904 ± 0.016.
- Three findings: window-track dip characterized; late-tail drain confirmed; complete 86-channel coverage and localization gate validated.

Backup: validation and gates

- 31/31 ROOT files pass: event IDs 0–1999, matching beam x, aggregate IDs 0–85.
- Top localization passes all positions: strict nearest maximum except the documented $x \equiv 10 \pmod{20}$ window-track effect (maximum among nearest two, deficit below 15%).
- At x=300 mm the two candidates differ by only 0.08 sigma and pass as a statistical tie.
- MPT checks pass: RINDEX 1.58, yield 10400/MeV, decay 1.8 ns, rise 0.7 ns, ABSLENGTH 160 cm, emission peak 408.8 nm.

Backup: ID18 impact maps



Backup: timing methodology

- SUM4 groups: {0..3}, {4..7}, {8..11}, {12..15}.
- Single-PE response: normalized 0.5/5 ns double exponential.
- Absolute leading-edge threshold: 4 PE equivalent.
- Estimator: $\sigma(\Delta T_{AB})/\sqrt{2}$.
- `HOOK_WALK`: pending; no time-walk correction exists or was invented.
- EXEC_09 anti-artifact audit verifies common threshold, pulse, t=0 and estimator.

Backup: glossary

HOOK_WALK Reserved insertion point for future leading-edge time-walk correction. Fixed 4-PE threshold means larger pulses cross earlier; no correction is applied in any campaign (`congruent_sum4_timing.C:305-315`).

FPT First photon time: first detected photon in an event for the stated channel/group.

SUM4 Analog sum of four SiPMs: {0–3}, {4–7}, {8–11}, {12–15}.

(exact geometry) Nearest geometric SiPM agrees with the maximum, or is covered by the documented window-track exception; x=300 mm is explicitly a statistical tie.

Temporal density Absolute plots are entries/event/ns; normalized plots integrate to one and compare shape only.

Central 24 mm pair The sole pitch exception is $-12/+12$ mm because 1384 mm is not divisible by 20 mm.

Intrinsic sigma Excludes SPTR and electronics jitter; the hardware 88 ps value is not directly comparable.

Backup: EndTop versus End-only tail audit